

PIZZA SALES ANALYSIS USING SQL



PROJECT OBJECTIVE:

HELLO I AM POOJA, AND IN THIS PROJECT I ANALYZED PIZZA SALES DATA TO UNCOVER VALUABLE INSIGHTS: REVENUE, PEAK SALES HOURS, CUSTOMER ORDERING TRENDS, CATEGORY-WISE SALES, TOP-PERFORMING PIZZA SIZES AND TYPES AND BUSINESS PERFORMANCE USING ADVANCE SQL QUERIES, ENABLING DATA-DRIVEN DECISIONS FOR BUSINESS GROWTH, INVENTORY PLANNING, AND OPERATIONAL EFFICIENCY.



ABOUT THE DATASET

Pizza Sales Analysis Project:

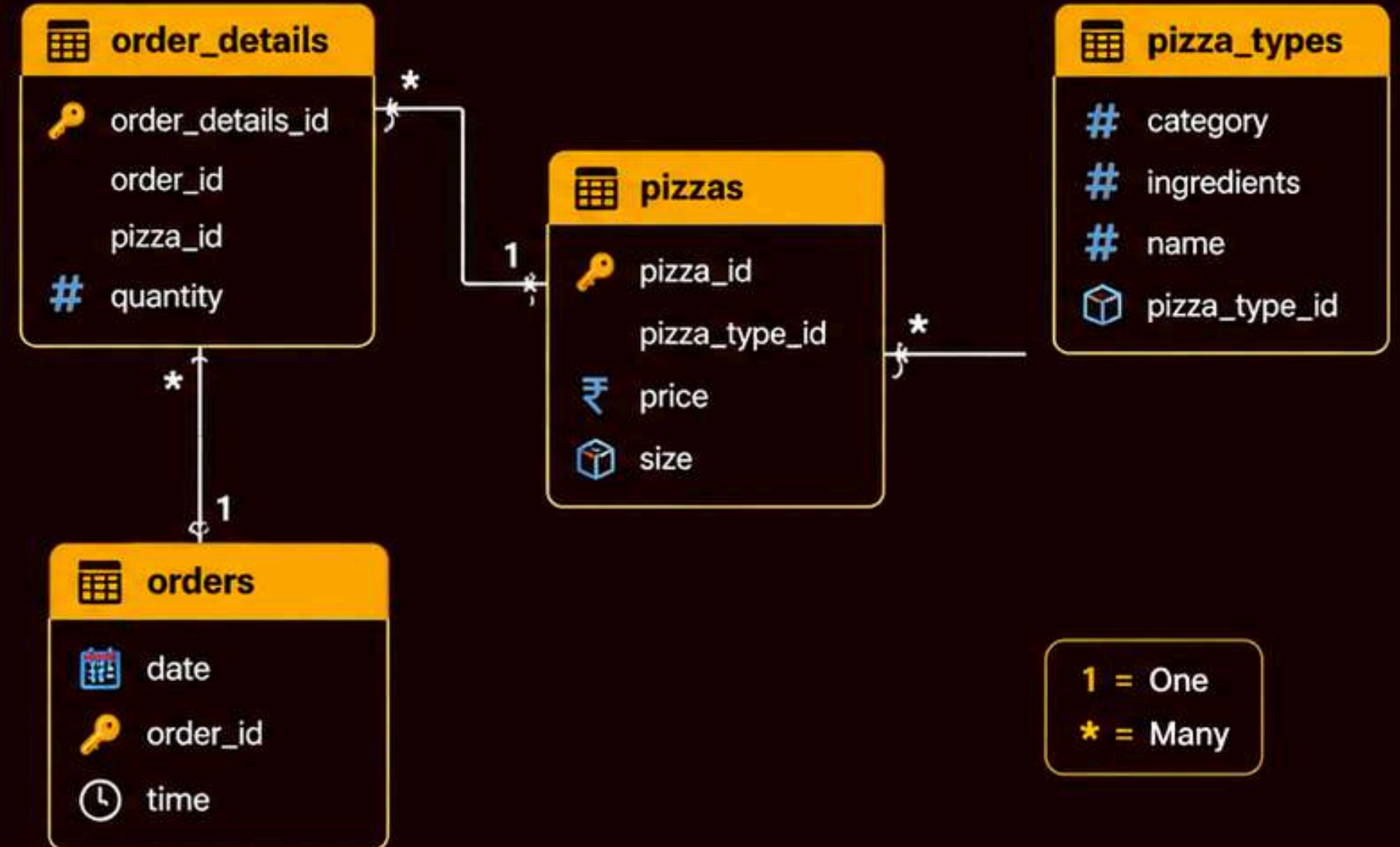
This project utilizes a relational pizza sales dataset with the following tables and key information.



TABLES INCLUDED

- Orders**
Contains order date and time details
- Order Details**
Stores quantity and pizza-wise order info
- Pizzas**
Includes pizza size and pricing details
- Pizza Types**
Contains pizza names, categories, and ingredients

DATABASE SCHEMA (ER DIAGRAM)



KEY COLUMNS

Order ID
Unique identifier for each order

Pizza Name
Name of the pizza ordered

Category
Type/category of pizza

Quantity
Number of pizzas ordered

Price
Price of each pizza

Order Date & Time
Date and time when order was placed

RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED

```
SELECT  
    COUNT(order_id) AS TOTAL_ORDER  
FROM  
    ORDERS;
```

Result Grid	
	TOTAL_ORDER
▶	21350

\$30



CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES

```
SELECT  
    ROUND(SUM(order_details.quantity * pizzas.price),  
          2) AS total_revenue  
FROM  
    order_details  
    JOIN  
    pizzas ON order_details.pizza_id = pizzas.pizza_id;
```



Result Grid	
	Total_Revenue
▶	817860.05

IDENTIFY THE HIGHEST-PRICED PIZZA

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```



Result Grid			Filter Rows
	name	price	
▶	The Greek Pizza	35.95	

IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS orders_count
FROM
    order_details
    JOIN
    pizzas ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizzas.size
ORDER BY orders_count DESC LIMIT 1;
```



Result Grid			Filter R
	size	orders_count	
▶	L	18526	

LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
        pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
        order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```



Result Grid   Filter Rows:		
	name	quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED

```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```



Result Grid			Filter
	category	quantity	
▶	Classic	14888	
	Supreme	11987	
	Veggie	11649	
	Chicken	11050	

DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY

```
SELECT
    HOUR(order_time), count(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time);
```

Result Grid			Filter Rows:
	HOUR(order_time)	order_count	
▶	11	1231	
	12	2520	
	13	2455	
	14	1472	
	15	1468	
	16	1920	
	17	2336	
	18	2399	
	19	2009	
	20	1642	
	21	1198	
	22	663	
	23	28	
	10	8	
	9	1	



JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```



Result Grid			Filter Rows:
	category	COUNT(name)	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	

GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY

```
SELECT
    ROUND(AVG(quantity), 0) AS avg_pizza_ordered_per_day
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```



Result Grid		Filter Rows:
	avg_pizza_ordered_per_day	
▶	138	

DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    order_details
    JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id
    JOIN
    pizza_types ON pizzas.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```



Result Grid			Filter Rows:
	name	revenue	
▶	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	

CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE

```
SELECT
    pizza_types.category,
    ROUND(SUM(order_details.quantity * pizzas.price)/(SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),2)
FROM
    order_details
    JOIN
    pizzas ON order_details.pizza_id = pizzas.pizza_id) * 100, 2) AS per_revenue
FROM
    order_details
    JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id
    JOIN
    pizza_types ON pizzas.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.category
ORDER BY per_revenue;
```



Result Grid			Filter Rows
	category	per_revenue	
▶	Veggie	23.68	
	Chicken	23.96	
	Supreme	25.46	
	Classic	26.91	

ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME

```
SELECT order_date, ROUND(SUM(Revenue) OVER (order by order_date),2) AS Cum_Revenue
FROM (select
  orders.order_date,
  SUM(order_details.quantity * pizzas.price) AS Revenue
FROM
  order_details
  JOIN pizzas ON order_details.pizza_id =pizzas.pizza_id
  JOIN
  orders
  ON orders.order_id = order_details.order_id
GROUP BY order_date) as Sales;
```



Result Grid		Filter Rows
	order_date	Cum_Revenue
▶	2015-01-01	2713.85
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.35
	2015-01-11	25862.65
	2015-01-12	27781.7
	2015-01-13	29831.3
	2015-01-14	32358.7
	2015-01-15	34343.5
	2015-01-16	36937.65
	2015-01-17	39001.75
	2015-01-18	40978.6
	2015-01-19	43365.75
	2015-01-20	45763.65
	2015-01-21	47804.2
	2015-01-22	50300.9
	2015-01-23	52724.6
	2015-01-24	55013.85
	2015-01-25	56631.4

Result Grid		Filter Rows
	order_date	Cum_Revenue
	2015-12-06	767549.45
	2015-12-07	769964.25
	2015-12-08	771820.5
	2015-12-09	774392.05
	2015-12-10	776377.65
	2015-12-11	779011.65
	2015-12-12	780971.8
	2015-12-13	783216.95
	2015-12-14	785389.55
	2015-12-15	787777
	2015-12-16	790011.8
	2015-12-17	791892.55
	2015-12-18	794778.85
	2015-12-19	797083.05
	2015-12-20	799187.95
	2015-12-21	801288.65
	2015-12-22	803171.6
	2015-12-23	805415.9
	2015-12-24	807553.75
	2015-12-26	809196.8
	2015-12-27	810615.8
	2015-12-28	812253
	2015-12-29	813606.25
	2015-12-30	814944.05
	2015-12-31	817860.05

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DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY

```
SELECT category, name, Revenue
FROM
(SELECT category, name, Revenue , rank() over(partition by category order by Revenue desc) as rn
FROM
(SELECT pizza_types.name, pizza_types.category, SUM(order_details.quantity*pizzas.price) AS Revenue
FROM
order_details
JOIN pizzas
ON order_details.pizza_id = pizzas.pizza_id
JOIN pizza_types
ON pizzas.pizza_type_id = pizza_types.pizza_type_id
GROUP BY pizza_types.category, pizza_types.name) as a) as b
where rn <= 3;
```

Result Grid			
Filter Rows:			
	category	name	Revenue
	Chicken	The Thai Chicken Pizza	43434.25
	Chicken	The Barbecue Chicken Pizza	42768
	Chicken	The California Chicken Pizza	41409.5
	Classic	The Classic Deluxe Pizza	38180.5
	Classic	The Hawaiian Pizza	32273.25
	Classic	The Pepperoni Pizza	30161.75
	Supreme	The Spicy Italian Pizza	34831.25
	Supreme	The Italian Supreme Pizza	33476.75
	Supreme	The Sicilian Pizza	30940.5
	Veggie	The Four Cheese Pizza	32265.7000
	Veggie	The Mexicana Pizza	26780.75
	Veggie	The Five Cheese Pizza	26066.5





THANK YOU